

HYPOTHESIS TESTING

****Module 2: The Shape of Chaos ****

1. THE TRIAL OF REALITY (H_0 vs. H_1)

In the mathematical language of the universe, **we assume nothing happened until proven otherwise**. When testing a new claim, you must set up two competing, mutually exclusive parallel realities:

- **The Null Hypothesis (H_0):** The baseline assumption of pure coincidence, zero effect, or the status quo.
 - *Example:* "This coin is totally fair." / "This new study-drug does not increase student exam scores."
- **The Alternative Hypothesis (H_1 or H_a):** The actual claim you are trying to prove.
 - *Example:* "This coin is rigged." / "This drug genuinely makes students score higher."

2. THE MATHEMATICAL SCALPEL (α)

Before you even look at your data, you have to draw a line in the sand. You have to answer the question: *"Mathematically speaking, how weird does a coincidence have to be before we refuse to believe it's a coincidence anymore?"*

This threshold is called your **Significance Level (α)**.

- The universal scientific standard is $\alpha = 0.05$ (a 5% threshold).
- By choosing $\alpha = 0.05$, you are officially declaring: *"I am willing to accept a 5% risk of accidentally declaring a true coincidence to be a real discovery."*

3. THE P-VALUE (THE MOST MISUNDERSTOOD NUMBER)

Once your experiment is finished, the computer hands you a decimal called a **P-value**.

The Golden Definition: Assuming the Null Hypothesis (H_0) is 100% true, the P-value is the exact probability of obtaining a result *at least as extreme* as the one you just measured.

- **The Coin Toss Analogy:** You flip a supposedly fair coin 10 times and get 10 Heads in a row. The P-value is the answer to the question: *"If this coin is genuinely fair, what were the odds of that happening by pure luck?"* (The math: $0.5^{10} = 0.00097$. Your P-value is 0.00097).

CRITICAL WARNING FOR STUDENTS: A P-value is **not** the probability that your hypothesis is correct. It is strictly a measurement of how surprised the Null Hypothesis is by your data.

4. SLICING THE MOUNTAIN (THE DECISION RULE)

To make your official scientific ruling, you take your calculated P-value and lay it next to your Significance Level (α). There are only two possible outcomes:

Scenario A: $P \leq \alpha$ (The Scalpel Cuts)

- **The Logic:** Your P-value returned 0.01 . Your cutoff was 0.05 . Because $0.01 \leq 0.05$, the data is officially "too weird" to be a standard Bell Curve accident.
- **The Scientific Move: Reject the Null Hypothesis (H_0).** Declare a Statistically Significant discovery.

Scenario B: $P > \alpha$ (The Blunt Edge)

- **The Logic:** Your P-value returned 0.18 . Your cutoff was 0.05 . Because $0.18 > 0.05$, there is an 18% chance this happened by pure Bell Curve bouncing. That isn't weird enough.

- **The Scientific Move: Fail to Reject the Null Hypothesis.**